

Modelling Interaction Dynamics for Behaviour Prediction in Motile, Physical Multi-Agent Systems

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I. INTRODUCTION

Multi-agent systems (MAS) are ubiquitous in the current computing landscape; these are systems comprised of multiple autonomous reasoning entities called ‘agents’ who interact within an environment, typically to accomplish individual or collective goals [1]. These systems have begun to eclipse traditional approaches in areas where distributed problem solving and autonomous decision making are increasingly useful, such as various logistic, infrastructure, and cyber security tasks [2] [3]. While LLM-based MASs are making their debut with frameworks like AutoGen, physical MASs have a rich history in robotics and autonomous systems; these are particularly important due to their deployment in safety-critical, real-world situations such as autonomous driving, coordinating drone swarms and visual recognition [4]. In settings where humans and agents interact, purely reactive behaviour is insufficient for maintaining safety and efficient decision-making, hence the ability to anticipate future actions accurately is necessary in order to engage safely and efficiently.

An example of this necessity is the advent of AVs being used in major cities, such as Waymo’s “Robotaxis”, and how there is a period of time in which these agents will share the roads with human drivers and pedestrians [5]. In order to become trustworthy participants to humans on the road, AVs must be able to understand the behaviours of human drivers and follow human-like principles so they can partake in a shared “driving language” and maximise cooperation between the two parties [6]. Providing agents with the ability to account for the likely trajectories of others can facilitate coordination, collision avoidance and alignment with human expectations, as such this is a fundamental problem for the reliable deployment of autonomous physical MASs in real-world environments.

Initially, this project focused on implementing a robotic swarm MAS using a *Multi-agent Reinforcement Learning* (MAREL) with a focus on using novel neural architectures. Despite the potential insights into the design of multi-agent policies in these specific environments, this potential project couples multiple complex factors such as reward design, policy exploration etc. This results in a difficulty to isolate interaction dynamics between agents, whereas behaviour prediction allows for a targeted investigation of these dynamics. Additionally, behaviour prediction models are crucial in real-world autonomous systems, operating alongside planning and control. Given these reasons, the project was redefined to focus on multi-agent behaviour prediction in order

to investigate interaction dynamics in a more interpretable manner.

This project investigates behaviour prediction in physical MASs with a focus on capturing interaction dynamics between agents. First, key challenges associated with predicting behaviour in these systems are identified and categorised. Based on these challenges, a machine learning approach is designed to model agent interactions and predict future behaviour in the form of trajectories. To evaluate this approach, an experimental pipeline is developed using both synthetic multi-agent simulations and the Stanford Drone Dataset as sources of labelled data. Through controlled experiments, the project aims to assess the significance of these challenges and to evaluate how effectively the proposed approach addresses them. Due to space constraints, all figures referenced throughout the report are presented in a dedicated figures appendix within the root of the project repository (<https://git.cs.bham.ac.uk/projects-2025-26/jxh1482>).

A. Objectives & Outcomes

In order to effectively carry out this task, a list of project objectives and learning objectives was created to guide the project.

- Learning Outcomes

- Investigate and identify key challenges involved in the prediction of multi-agent behaviour, particularly within physical environments.
- Analyse and categorise these challenges into a set of problem groups that can guide the design and experimentation of the model
- Evaluate the effectiveness of a proposed modelling approach through controlled experiments to assess how well it addresses the identified challenges.

- Project Objectives

- Select a neural architecture for modelling multi-agent interactions, with model design choices justified based on their effectiveness in addressing identified challenges.
- Design and implement a machine learning approach capable of predicting future trajectories within synthetic and real-world datasets.
- Build an experimental pipeline using both synthetic multi-agent simulations and real trajectory data to train and evaluate predictive models.

These learning objectives were used to synthesise a list of research questions to form the basis of the following literature

review. The resulting investigation into these question areas, and the insights gathered are repurposed for key decision-making in execution of the project objectives.

- **Research Questions**
 - What are the challenges in performing multi-agent behaviour prediction in physical MASs?
 - In the literature, how do different proposals approach the problem and the identified challenge areas?
 - For a candidate approach to this problem, what design choices can be inferred from the literature?

II. RELATED WORK & LITERATURE REVIEW

This Literature Review examines existing research on behaviour prediction in physical MASs. The aim is to answer the aforementioned research questions by identifying and categorising the key challenges associated with predicting multi-agent behaviour, analysing approaches proposed in the literature, reviewing how these challenges have been addressed, and uncovering potential ways these approaches could influence the one proposed in this project.

A. Search Methodology

The goal of this Search Strategy is to retrieve relevant research papers and other literature mediums related to this topic. Sources are selected using the following process:

- **Databases:** Papers were sourced from arXiv (Primary), IEEEExplore, ResearchGate, ScienceDirect and the ACM Digital Library.
- **Search Terms:** The following Search Terms were combined using boolean operators to construct search strings to explore the databases: *Multi-agent, Interaction, DNN, GNN, RNN, MLP, Graph, Supervised Learning, Reinforcement Learning, Swarm, Autonomous Agents, Robots, Dynamic System, Behaviour Prediction, Trajectory Prediction, Autoencoder, Adversarial, Cooperative, Convolution.*
- **Selection Criteria:** Sources were included if they provided relevant insight into MASs, trajectory prediction, or neural architectures. Sources were excluded if they were not published within the last decade in order to maintain relevancy to the modern state of the field; exclusions were made for sources that are particularly relevant to this specific task or foundational work referenced by more recent sources.
- **Search Strategy:** A backwards snowballing method was utilised in which initial “seed” sources closely aligned with the research topic were identified and references within these papers were then examined. Relevant sources that expand on their methodologies, results, or theoretical foundations are located and the process is repeated iteratively to build a bibliography.

Guided by these questions and methodology, the relevant literature has been surveyed to derive answers in the following subsection.

B. Research Synthesis

1) *What are the Challenges in Performing Multi-agent Behaviour Prediction in Physical MASs?:* When observing a physical MAS, typically only individual agent trajectories are available, leaving the knowledge of underlying dynamics and patterns between agents unknown [7]. The literature suggests that given agent behaviour can be influenced by multiple factors, the same observed past behaviour may correspond to multiple plausible futures that exhibit distinct trends: Wu & Ying-Ching [8] refer to this as the “multi-modality problem”. Furthermore, traditional approaches such as “Kalman Filters” and “Time-series Analysis” have been deemed insufficient for multi-agent behaviour prediction when a given scenario is sufficiently complex and/or agents do not behave independently [9]. Given this prevalent issue, it’s been suggested that “learning-based models” with large capacities may be required to capture the dynamics which make multiple futures difficult to model [9]. Failing to address these different modalities leaves an approach susceptible to “mode collapse” in which a model fails to capture the full diversity of potential outputs and over-optimises towards a single mode, destroying model performance [10]. It’s clear from the literature that ‘Multi-modality’ is a prevalent challenge area for behaviour prediction, particularly in multi-agent environments such as physical MASs where agent behaviour is inter-dependent and key dynamics are unknown.

It can be assumed that the behaviour of highly interactive multi-agent systems are dependent on the agents themselves and the relationships/dependencies between agents. Given this interconnectivity, Li et al. [9] suggest that modelling agents as individuals or an entire system through an “ego” agent fails to capture the interaction dynamics most representative of future behaviour and instead propose treating multiple agents as a singular, whole system in order to “model the joint distribution of their future behaviours”. The nature of these agent interactions is expanded upon: It’s put forward that these underlying interactions are “inherently varying” due to abrupt changes in context and by heterogeneous types of agents within a MAS [10]. It’s stressed that capturing the dynamics of interaction relies on handling these heterogeneous agents and interactions in a “unified way”, corroborating the previous source [10]. The literature is clear that modelling interaction dynamics between agents is a key challenge area that requires a pervasive approach which accounts for shifting contexts, heterogeneous agents and varying interactions in order to capture these dynamics, leading to more accurate predictions.

Looking beyond the current topology of a MAS into the previous behaviours of agents through time provides a new dimension in which to extract information about a system’s behaviour and consequently it’s future actions. Wu & Ying-Ching [8] highlight the different requirements for temporal modelling in MASs at different scales: Short-term predictions using small temporal frames are reported to rely on local predictions and fine-grained/immediate motion dynamics, conversely long-term predictions using large temporal frames are reported to rely on global predictions and larger behavioural trends. These competing priorities create

a challenge in modelling temporal dependencies because an individual model attempting to accurately predict behaviour in both contexts will need to make trade-offs in its design which may limit its effectiveness in any individual instance. When implementing temporal modelling, recorded system states of a MAS can be represented as time series data, which lends itself naturally to sequential/recurrent/auto-regressive architectures including RNNs/LSTMs, GRUs and repeated use of generator modules [8, 10, 9]. A fundamental issue in using this class of architecture for temporal modelling in a MAS is “cumulative prediction errors” where small inaccuracies in previous steps compound and amplify inaccuracies in following steps [8]. It has been argued this issue is innate to behaviour prediction with these architectures as a consequence of the uncertainty in agent behaviour increasing substantially at higher prediction horizons, resulting in a degradation of long-term temporal modelling performance [8]. Existing research presents temporal modelling as a key challenge area in physical MAS behaviour prediction; the handling of short-term and long-term dependencies in addition to innate challenges with certain architectures both have the potential to strongly impact the performance of any proposed approach.

In most contexts, any prediction model which has the intention of being used in real-world scenarios should have the capacity to generalise to new and unseen situations because real-world data is inherently complex, noisy, and constantly evolving, hence unlikely to be directly encountered before. This is particularly relevant in physical multi-agent scenarios, such as autonomous driving, where “domain shift” is inevitable [4]. The literature regards this as an open challenge for learning-based approaches on the basis that appropriate datasets are difficult to obtain and at times do not match the operating conditions likely to be encountered [7]. It’s claimed that this challenge in making generalisable approaches may be worsened due to a particular focus on improving overall prediction performance in the literature at the expense of focusing on generalisation [11]. It’s clear from these sources that although the ability to generalise is important for successful multi-agent behaviour prediction in unseen scenarios, its specific implementation in this task is under resourced and a gap in the research.

In this report, there have been a large number of individual challenges identified in behaviour prediction for physical MASs. Among the challenges reviewed, they can be filed into four key challenge areas: **Multi-modal** challenges, **Interaction Modelling** challenges, **Temporal Modelling** challenges, and **Generalisation** challenges.

2) *In the literature, how do different proposals approach the problem and the identified challenge areas?:* The literature contains various methods for dealing with the identified challenge areas using unique and occasionally contrasting methods (Summarised in Table 1).

In temporal modelling, Wu & Ying-Ching [8] attempt to eliminate cumulative prediction errors by leveraging *Convolutional Neural Networks* (CNNs) from computer vision to capture latent motion dynamics by encoding all historical data in a single pass, bypassing the conditions for compounding inaccuracies and reducing latency in real-world

use. This technique is further developed by identifying an endpoint before decoding the latent motion dynamics into a complete trajectory to reduce any uncertainty in predicting the sequence. Contrasting this approach, Li et al. [10] model temporal dependencies by re-encoding a latent interaction graph every τ time steps using *Gated Recurrent Units* (GRUs). Although this approach may be susceptible to cumulative prediction errors, re-encoding in real-time allows for historical information to be propagated forward and allow the modelling of shifting interaction dynamics over time. While neither approach is strictly advantageous, these demonstrate how different priorities can affect the design of a behaviour prediction model and that there may be inherent trade-offs when designing temporal modelling.

For the multi-modality problem, *Gaussian Mixture Models* (GMMs) have the potential to produce multiple potential futures from a singular input [10]. This is done through outputting the weights, means and variances for K Gaussian components representing distinct action “modes” [10]. This specific implementation uses a “Minimal Loss Hypothesis” in which only the one of δ potential trajectories output during training which has the lowest loss w.r.t. the ground truth is used in backpropagation; this is done to encourage feasible trajectories while avoiding “mode collapse” [10]. Probabilistic approaches appear in the literature more than once, for example a *Generative Adversarial Network* (GAN) consisting of an opposing generator and discriminator have been used to force a model to learn a distribution over all possible future system states [9]. When accounting for multi-modality, the literature appears to converge on probabilistic approaches, albeit with different methods of engaging with the variety of future system states.

Agents in a physical MAS can operate under diverse policies and physical constraints, which may be either homogeneous or heterogeneous across the system. It appears that these characteristics manifest the emergent complexity of multi-agent behaviour and the difficulty in modelling these interaction dynamics. Using graph-based models to represent the interactive and physical topologies of physical MASs appears to be prevalent in the literature [7, 10, 11]. These graph-based models leverage a message-passing mechanism to explicitly model interactions between agents and produce approximations of emergent multi-agent behaviour through node-level regression [12]. These graph-based models are able to be extended to more comprehensively capture interaction dynamics between agents. For instance, separate, specialised role encoders have been used to more effectively model heterogeneous agent roles (e.g. cars & pedestrians in a driving context) than using a universal embedding function [10]. Additionally, giving agents distinct “self” (immutable) and “social” (mutable through interaction) attributes has been demonstrated to improve modelling of shifting social dynamics [10]. The research indicates that utilising graph-based models for behaviour prediction in physical MASs is an extensible and established approach for handling interaction modelling.

It’s evident these encoders are vital for both interaction and temporal modelling due to their prevalence in the literature. Van der Hoek & Wooldridge [1] use ideas from

the “philosophy of mind” to elaborate on what these encoders may be doing on a macroscopic level, this source maps the “BDI [Belief-Desire-Intention] theory of human rational action” proposed by Michael Bratman onto rational agents within a MAS. These encoders can be thought to be deriving an agent’s BDI model from contextual information which effectively filters the reasoning for future behaviour.

A key weakness of multi-agent behaviour prediction models seems to be generalising to different tasks/layouts they haven’t been directing trained on. In contrast to using Cartesian coordinates as input for the current topology of a physical MAS, Hengbo et al. [11] use self-supervised learning to extract domain knowledge and produce a “Frenét-based representation”. This is a reference coordinate system which explicitly incorporates scenario information (in this case, road layouts for AVs) into the feature space and improving zero-shot transferability by “30.8%” when compared to an equivalent Cartesian-based model [11]. Despite light coverage in the literature, the problem of generalisation can be addressed through altering how information is represented within a predictive approach.

In this report, the four key challenge areas have been approached in various ways by proposals in the literature. **Multi-modal** futures were shown to be expressible using probabilistic approaches, **Interaction Modelling & Temporal Modelling** could be performed by using different architectural paradigms, and **Generalisation**, although lightly addressed, could still be imparted by altering data representation.

3) *For a candidate approach to this problem, what design choices can be inferred from the literature?:*

It is clear that *Graph Neural Networks* (GNNs) are uniquely suited to modelling a physical MAS based on repeated occurrences in the literature [7, 10, 11]. Agents within such a system disseminate information to others, whether explicitly (messaging/communication) or implicitly (observing other agent’s actions/properties). This process is heavily mimicked by a GNN’s message-passing framework where nodes exchange vector messages that diffuses feature information among all connected agents in the topology [7]. Given this congruence, GNNs are able to be effectively used to model the evolving dynamics of agent states (node attributes) and agent relationships/interactions (edge attributes) [10]. Moreover, GNNs may aid generalisation to varied physical MAS scenarios of different structures and sizes; Fallin et al. [7] attribute this to the paradigm being “flexible and generic” which may be a key to robust behaviour prediction frameworks. While alternative architectures have been explored in the literature, they are often used as complementary components rather than as the core modelling framework. For example, hybrid approaches combine GNNs with temporal encoders or domain-specific representations [10, 11]. In contrast, architectures such as *Recurrent Neural Networks* (RNNs), GANs, and transformers present limitations when considered as the primary modelling approach for this task. Recurrent models struggle to capture long-range dependencies due to issues such as vanishing/exploding gradients, while adversarial models, although capable of modelling multi-modal outputs, are often difficult to train

and lack stability [9, 8]. Transformer-based approaches offer a strong representational capacity but introduce significant computational overhead and may be prone to over-fitting in settings with limited or sparse data [8]. These observations suggest that while such architectures may be beneficial as supporting components, GNNs provide a more suitable foundation for modelling multi-agent interaction dynamics due to their ability to explicitly represent relational structure, combined with their flexibility across different system topologies.

Capturing long-term dependencies and distilling an agent’s “BDI state” (Van der Hoek & Wooldridge [1]) may require using a temporal encoder to process an agent’s features over time into a latent embedding, evidenced by this technique’s prevalence in the research [8, 10, 9]. These are often paired with decoders which parse these latent embeddings into predicted trajectories and construct a sequence of future system states [8, 10, 9]. For this project, using a single pass encoder/decoder structure could be effective in avoiding cumulative errors altogether and reducing latency (significant in real-time physical MAS contexts like autonomous driving) [8]. This decision is a compromise with the trade-off being not using iterative/recurrent probabilistic methods to model multi-modal futures (e.g. GMMs, GANs) [10, 9].

Effective evaluation of multi-agent behaviour prediction models requires both appropriate error metrics and representative datasets that reflect the complexity of physical MAS. The literature consistently adopts *Average Displacement Error* (ADE) and *Final Displacement Error* (FDE) as standard metrics for trajectory prediction, as they capture both overall alignment with ground truth trajectories and performance at longer prediction horizons [10]. Variants such as *Non-linear ADE* (NL-ADE) further refine this evaluation by isolating performance on complex, non-linear motion patterns which allow models to be assessed beyond simple or near-linear trajectories [13]. In addition to metric selection, the choice of training and evaluation data plays a critical role. Existing work highlights the importance of capturing both short-term and long-term temporal dynamics, which are often under-represented within a single dataset [8]. Synthetic data generated from controlled multi-agent simulations could provide abundant and diverse examples of fine-grained, short-term interactions, while real-world datasets such as the *Stanford Drone Dataset* (SDD) could offer more complex, long-horizon behaviours shaped by scene context and agent intent [8, 14]. These observations imply that a robust evaluation framework should combine complementary datasets and multiple error metrics to fully assess an approach’s performance across different temporal scales and behavioural complexities. This informs the design of the experimental pipeline in this report, where both synthetic and real-world data are utilised alongside standard trajectory prediction metrics to provide a comprehensive evaluation.

Many design choices for a candidate approach can be derived from and informed by the existing research. A graph-based model, combined with single pass encoders and decoders, and trained on a combination of synthetic and

real-world multi-agent trajectory data provides a sweeping framework for multi-agent behavior prediction in physical MASs. Despite trade-offs in **Multi-modal** modelling, this candidate architecture thoroughly addresses **Interaction Modelling**, **Temporal Modelling** and **Generalisation**.

C. Summary

This literature review examined existing research on behaviour prediction in physical MASs in order to identify key challenges and inform the design of an appropriate modelling approach. The sources consistently highlight several major challenge areas associated with this task, including multi-modality in possible future behaviours, complex interaction dynamics between heterogeneous agents, modelling temporal dependencies across different prediction horizons, and limited generalisation due to sparse or domain-specific datasets.

Existing approaches address these challenges through a variety of architectural/modelling strategies, including probabilistic trajectory modelling, adversarial training, temporal encoding mechanisms, and explicit modelling of agent interactions. Within these methods, graph-based representations are frequently used to capture the relational structure between agents and evolving interaction patterns within the system.

Based on this research, GNNs emerge as a particularly suitable architecture for modelling behaviour prediction in physical MAS. Their message-passing framework enables agents to exchange contextual information while naturally representing the topology of interactions amongst agents. Combined with appropriate temporal encoders and trajectory decoders, such architectures provide flexibility for modelling multi-agent dynamics while addressing the key challenges identified in the literature. These insights inform the design of the modelling approach implemented in this project and guide the system design and experimental evaluation described in the following sections.

III. DESIGN & IMPLEMENTATION

A. System Overview

The project's goal is to design and implement an approach to multi-agent behaviour prediction in physical contexts and then evaluate the approach through controlled experiments and data analysis. To accomplish this, an experimental pipeline was designed. Multi-agent simulations generated trajectories in various contexts, raw trajectory data is processed into supervised training examples, graph datasets are constructed from these examples for training and testing, outlined models are trained and evaluated, and experiments are conducted to test design features.

Core components of this pipeline include a file system data store to index raw data, constructed datasets and trained models for future use, a simulation platform to generate raw data, a hyper-parameter store to vary function of different pipeline components, and finally a collection of scripts which act as core processing components and as a human interface to the system. This pipeline is represented as a flowchart in Figure 1.

B. Behaviour Prediction Model

1) *Input Representation*: The produced approach uses a series of consecutive 2D 'snapshots' of the MAS as the input, each 'snapshot' contains a graph-based representation of the MAS at a given time-step in which agents/landmarks are nodes. Nodes consist of a list of features separated into "static" features (position stored as Cartesian coordinates) and "social" features (agent role, velocities and orientation) derived from observable information. The former are used to construct edges between nodes based on proximity using K-Nearest Neighbours with each edge containing three features (displacement in each axis and the Euclidean distance). This approach was implemented using PyTorch Geometric so each input is encoded as a PyG 'Data' object [15].

2) *Temporal Encoding*: Historical context or "burn-in" is used to create a per-agent embedding representing an agent's motion dynamics and BDI Model in the current temporal frame. This temporal encoder is implemented as a *Multi-layer Perceptron* (MLP) which takes a tensor containing each agent's "social" features over each historic time-step in the input and outputs a tensor containing a fixed sized embedding for each agent. The MLP encoder contains a fixed-size hidden layer between the input and output with non-final layers containing ReLU activation functions to introduce non-linear feature extraction. Using this format allows for every node's historical context to be encoded in a single-pass and eliminate latency linked to the number of agents of size of temporal frame.

Edges are encoded using an MLP of the same structure.

3) *Interaction Modelling*: Interactions between agents are modelled using message-passing layers. Within layer k , each node i produces a message $m_{i,j}^{(k)}$ for each connected node $j \in N(i)$ in the graph as a function of both nodes' embeddings $(h_i^{(k)}, h_j^{(k)})$ and the embedding of the connecting edge $e_{i,j}$ (edges in this approach are bidirectional given they are position based). Each node aggregates their incoming messages $m_{j,i}^{(k)}$ using a given operator and then create an updated embedding $h_i^{(k+1)}$ as a function of their previous embedding $h_i^{(k)}$ and their aggregated messages $a_i^{(k)}$.

$$\begin{aligned} h_i^{(k+1)} &= \text{UPD}(h_i^{(k)}, \text{AGG}(\{m_{j,i}^{(k)} \mid \forall j \in N(i)\})) \\ m_{i,j}^{(k)} &= \text{MSG}(h_i^{(k)}, h_j^{(k)}, e_{i,j}) \end{aligned}$$

In this specific implementation, the *message* (MSG) and *update* (UPD) functions use MLPs. The intention is that training will result in these functions learning what information is important to be communicated and how to most effectively integrate this information into an agent's embedding to model interaction between agents and consequently improve behaviour prediction performance. The *aggregation* (AGG) function use summation of every incoming message. Using message-passing allows this model to exhibit permutation invariance and equivariance provided by the message creation and aggregation functions respectively [16].

Source	Core Architecture	Key Challenge Addressed	Notable Technique
J. Li et al, 2019 [9]	GAN-based model	Multi-modality	Adversarial trajectory generation
J. Li et al, 2020 (EvolveGraph)[10]	GNN + GRU	Interaction dynamics, multi-modality	Dynamic latent interaction graphs
H. Ma et al, 2021 [11]	GNN-based model	Generalisation	Frenet coordinate representation
B. C. Fallin et al, 2025 [7]	GNN framework	Interaction modelling	Message-passing representation
J. W. Wu et al, 2026 (CoordConv) [8]	CNN encoder-decoder	Temporal modelling	Single-pass trajectory prediction

TABLE I: Comparison of Key Approaches to Multi-agent Behaviour Prediction.

$$h_i^{(k+1)} = h_i^{(k)} + \text{MLP}_{\text{UPD}}\left(\sum_{j \in N(i)} m_{j,i}^{(k)}\right)$$

$$m_{i,j}^{(k)} = \text{MLP}_{\text{MSG}}(h_i^{(k)} \parallel h_j^{(k)} \parallel e_{i,j})$$

This approach uses three message-passing layers which is within the acceptable range [1..4] for “most state-of-the-art GNNs” [17]. This means that any pair of nodes within three connections of each other will impact the embedding of the other, explicitly modelling how interactions between agents can impact more than just the direct participants.

This behaviour prediction approach is demonstrated architecturally in Figure 2.

C. Multi-agent Environment Simulation

1) *Simulator Selection*: Selecting an appropriate simulation platform involved creating a list of criteria to evaluate how closely each candidate simulator aligns with this project’s goals and directly compare them to each other. The criteria was organised from most to least important: **Multi-agent Support, Scenario Ecosystem, Performance, Customisation Potential, Ability to Log, Physics Fidelity and Friction of Use**. A more thorough explanation of the criteria, the candidate simulators and the selection process is summarised in Figure 3.

Vectorised Multi-agent Simulator (VMAS) [18] became the obvious choice, despite low physical fidelity, for its flexible, modular framework purpose-built for efficient multi-agent simulation. VMAS uses vectorised *Multi Particle Environments* (MPEs) to simulate multiple scenarios and multiple agents per scenario in parallel while providing an easy-to-use interface to collect observations about agents. In addition, VMAS provides a number of “scenario” environments as default which, when paired with enhanced ports of MPE agent controllers, allows for the generation of massive amounts of synthetic MAS behaviour data that can be highly tuned towards specific experimental goals. A selection of these default scenarios are shown in Figure 4.

2) *Scenario Design*: Six scenarios from the VMAS library were chosen for implementation, five of which were implemented and four used in experiments. These scenarios were chosen because they exhibit different forms of interaction dynamics, decision ambiguity, and temporal coordination between agents. Collectively, they provide controlled environments that expose the approach to varying manifestations of the key challenge areas.

The chosen, implemented and used scenarios consist of the following.

- **Flocking**: Agents adjust their velocities and headings based on neighbouring agents, producing coordinated group movement. This scenario evaluates the approach’s ability to capture emergent collective behaviour determined by local interaction rules.
- **Discovery/Navigation**: Agents choose and navigate across a possible route to reach a goal depending on local conditions and interactions with other agents. This scenario evaluates whether the approach is able to predict future behaviour in situations where multiple feasible futures could occur.
- **Football**: Agents operate in competing teams to pursue opposing objectives in a simulated football game. This scenario evaluates heterogeneity in multi-agent dynamics, requiring the approach to model both cooperative and adversarial interactions in the latent interaction graph.
- **Simple Tag**: Agents engage in a pursuit-evasion task where predators pursue prey in an obstacle populated environment. This scenario evaluates the approach’s fine-grained interaction modelling and short-term prediction ability due to highly reactive behaviour leading to rapid directional changes.

The **Transport** scenario was initially implemented because it introduced cooperative behaviour between agents manipulating a shared object. However, this relies heavily on physics-based interactions which introduced instability in simulation outputs during large-scale data generation. As a result, this scenario was excluded from experimental evaluation. The **Passage** scenario was considered during the design phase but was ultimately not implemented due to its similarity to other selected scenarios, particularly those involving constrained navigation and agent avoidance behaviours. Given the goal of exposing the approach to diverse situations, it was unlikely this scenario would produce novel results.

3) *Agent Controllers*: For each implemented scenario, vectorised agent controllers were designed to provide the scenario environments with activity. **Football** utilised a native VMAS controller whereas the other four implemented scenarios used vectorised ports of existing MPE controllers. In some cases these were modified to provide more complex behaviour. Controllers were used instead of *Deep Learning* (DL) or *Reinforcement Learning* (RL) because they offer high stability, interpretability and lower data requirements.

D. Data Processing Pipeline

1) *Raw Simulation Data*: Multi-agent simulations were run using VMAS and the data generation interface. Individual runs

were configured with 'scenario', 'number of environments', 'number of time steps' per environment and 'number of agents' per environment. Data recorded from these runs was saved as CSV time series data using a set schema: This included a composite key to identify individual agent frames in a specific run in addition to positions, velocities, roles and orientations filed into 'static' and 'social' features (detailed in Table 2).

2) *Dataset Construction*: Raw data is transformed into distinct supervised training and testing datasets using the data processing interface. Data processing is configured with 'history', also known as "burn-in", 'horizon', 'number of neighbours' per agent, 'minimum agents' per example and 'ratio' of training to testing examples. The raw data is grouped by distinct simulation run ("scene") and then a sliding window approach is used to aggregate agent data within a given time frame anchored around a specific time step; the number of time steps before the anchor is specified by 'history' which quantifies the historical context given to the model and the number of time steps after the anchor is specified by 'horizon' which provides the prediction goal of the model. Time series data within the window is converted into an individual example with agent histories, current positions and future ground truth relative positions.

These examples are further processed into PyG 'Data' objects by specifying nodes with each agent's history and future positions and using a K Nearest Neighbours approach to create connections between nodes based on current positions. These examples are split between training and testing datasets according to the 'ratio'.

E. Training Framework

1) *Model Training*: Training instances of a given model/approach is done using the training interface. This process is configured with 'model type', 'number of epochs', 'batch size' of examples used in a single forward/backward training pass and the 'encoding dimension' of the temporal and edge encodings in the approach. A specified training and testing dataset pair is loaded from the pipeline's datastore and a unique instance of the approach is created with specified dimensions. The approach is trained in batches within each epoch using *Adaptive Moment Estimation* (Adam) over other optimisers (e.g. SGD) because it's second-moment normalisation mechanism provides faster, more robust convergence with less need for tampering with the learning rate [19].

2) *Model Evaluation*: Following each training epoch, the approach is exposed to the testing dataset and its MSE Loss across the entire testing dataset is calculated. Despite being poor at representing multi-modal outcomes, MSE provides stability in gradient-based optimisers and is a powerful baseline metric for short-term accuracy. The training framework keeps track of the lowest overall test loss observed for a given epoch in order to perpetually keep a copy of the best performing model weights. This 'checkpointing' mechanism allows training approaches with high epochs without risking decreased performance in the output model from overtraining.

F. Ablation Models

The approach proposed in this report contains two major architectural features (message-passing, temporal encoder) which address two major challenge areas directly (interaction modelling, temporal modelling). In order to evaluate the effectiveness of these features and the overall impact these challenge areas impose on behaviour prediction in a physical MAS, variants of the approach have also been designed which lack one of or both of these features to conduct an ablation study. A summary of these models and the implementation is shown in Table 3 and Figure 5 respectively.

G. External Dataset Integration

Although the fine-grained motion dynamics that are key to evaluating short-term behaviour prediction were provided through simulation and synthetic data, long-term realistic physical MAS behaviour is difficult to simulate while avoiding cumulative errors without extensive resources. To evaluate model performance on this subset of physical MAS behaviour the Stanford Drone Dataset (SDD), a large-scale human trajectory dataset where various agents navigate in a real world outdoor environment, was integrated into the existing project pipeline [14].

This involved a pre-processing stage where SDD video annotations were combined for each 'scenario' and the data mapped on to the existing raw data schema. Various guarantees of synthetic scenarios (e.g. trajectory continuity, no occlusions, no interpolations from annotated data) weren't present in the external data so measures such as carrying forward previous data to fill gaps, voiding occluded data points and calculating missing features from existing features were utilised. Images of the real-world scenarios selected from SDD for this project are available in Figure 6.

One issue encountered in pre-processing SDD data was that while the videos (and annotations) were captured at 30 frames per second, changes in agent position were only updates every few frames. 30 fps sampling was too frequent for this project's purposes so multiple frames of agent trajectory information were compressed into a single entry in the produced raw data. This provided a unique benefit because the ratio of compression to real world time elapsed could be configured, allowing for variance in the length of time steps extracted from the external dataset (e.g. 0.2s, 0.5s, 1s per time steps).

IV. EXPERIMENTS

A. Experiments Overview

Following implementation, the proposed behaviour prediction approach was evaluated through a series of experiments designed to investigate two of the primary challenges identified in the literature: Temporal modelling and interaction modelling. While the broader problem also involves challenges such as multi-modality and generalisation, the approach specifically targets the modelling of temporal dependencies and agent interactions through the use of historical context encoding and graph-based message passing.

Hence, two experiments were designed to analyse the impact of these components. The first experiment investigates

Composite Key			Static Features		Social Feature: Agent Role (One-Hot-Encoding)										Social Features		
sceneId	timestep	agentId	posX	posY	role1	...	...	...	...	...	...	...	...	role11	velX	velY	heading

TABLE II: Raw CSV Time Series Data Schema

Model Variant	Removed Component
Extended GNN	None
Baseline GNN	Temporal Encoder
Extended MLP	Message-passing
Baseline MLP	Temporal Encoder & Message-passing

TABLE III: Model Variants for Ablation Study

the role of temporal context by analysing how varying the historical trajectory information provided to the model influences prediction accuracy across different horizons. The second experiment evaluates interaction modelling by examining how the connectivity of the input graph affects predictive performance. Together, these experiments aim to provide insight into how effectively the proposed solution captures temporal dependencies and interaction dynamics within physical MAS.

B. Experimental Setup

1) *Datasets & Scenarios*: A number of simulated and real-world scenarios are used in these experiments to evaluate how the approach may generalise to different contexts; the variation between scenarios allows more nuanced findings in approach performance to be expressed. In these experiments, the **Flocking**, **Navigation**, **Football**, and **Simple Tag** simulations as well as the **Coupa** and **Little SDD** scenario annotations were used to produce training and testing data.

2) *Model Variants*: For the purpose of quantifying the impact of the tested components, these experiments use variants of the approach that lack the critical components and are tested in the same conditions. This allows the experiments to double as an ablation study. Both experiments involve the **Extended GNN** with full functionality and each experiment utilises the **Baseline GNN** and **Extended MLP** respectively.

3) *Training Procedure*: Different instances of raw data and processed datasets were generated/sourced for each experiment as both require different variables to be controlled and others varied. Across both experiments, the training/testing ratio used is 4:1 and every model was trained using the checkpointing method described previously with a maximum of 30 epochs and a batch size of 16 examples per optimisation step. Every model instance trained was given an encoding dimension of 128 which governs the size of the temporal/edge encodings and the size of hidden layers in all utilised MLPs, this was done to eliminate performance improvements in models/model types purely based on number of parameters. For external datasets, the minimum number of agents per example was set at two. This is a trade-off which simultaneously prevents interactions being entirely eliminated, while maximising the training/testing examples that could be extracted from a dataset.

4) *Evaluation Metrics*: Prediction performance was evaluated using standard trajectory prediction metrics. *Average Displacement Error* (ADE) is computed as the average displacement error between predicted and ground truth trajectories, accumulated over time and normalised per time step. *Final Displacement Error* (FDE) measures the distance between predicted and ground truth trajectories at the endpoint of the prediction horizon (T) only. In addition, *Non-Linear Average Displacement Error* (NL-ADE) was used to specifically evaluate prediction accuracy on non-linear trajectory segments, providing insight into approach performance on more complex motion [20].

$$\begin{aligned} \text{ADE}^{(T)} &= \frac{\sum_{i=1}^n \sum_{t=1}^T \sqrt{(\hat{x}_i^t - x_i^t)^2 + (\hat{y}_i^t - y_i^t)^2}}{n(T)} \\ \text{FDE}^{(T)} &= \frac{\sum_{i=1}^n \sqrt{(\hat{x}_i^T - x_i^T)^2 + (\hat{y}_i^T - y_i^T)^2}}{n} \\ \text{NL-ADE}^{(T)} &= \frac{\sum_{i=1}^n \sum_{t=1}^T I(x_i^t, y_i^t) \sqrt{(\hat{x}_i^t - x_i^t)^2 + (\hat{y}_i^t - y_i^t)^2}}{\sum_{i=1}^n \sum_{t=1}^T I(x_i^t, y_i^t)} \end{aligned}$$

$$\text{where } I(x_i^t, y_i^t) = \begin{cases} 1 & \text{if segment is non-linear} \\ 0 & \text{otherwise} \end{cases}$$

C. Experiment 1, Temporal Modelling

This experiment's goal is to test how varying the historical context provided to the approach can affect its ability to predict future trajectories in a physical MAS.

1) *Design*: To assess the temporal modelling, the number of historical time steps provided in an agent's initial features was varied. Five history lengths were evaluated: 1, 3, 5, 10, and 20 time steps. For each configuration, models were trained and evaluated across prediction horizons ranging from 1 to 20 time steps. This process was repeated using both the **Extended GNN** and **Baseline GNN** and was separately conducted across the flocking, navigation, and coupa (SDD) scenarios. For every instance in this experiment connectivity was kept static with each agent having two neighbours. For each simulation, 64 independent instances were run, each containing 8 agents and running for 300 time steps; this was done so that the diversity within each scenario could be trained on and evaluated.

2) *Hypothesis*: The hypothesis is that providing more historical context will improve the model's prediction performance and that this will be an ongoing trend with diminishing returns assuming that immediate history is more representative of an agent's BDI state than older history. It's expected that the model variant lacking a temporal encoder will perform worse given it's inability to encode agent features into a high-level representation. It's expected that shifts in performance will be more noticeable in the navigation scenario

than the flocking scenario given the former relies on historical information to identify the direction of an agent’s goal position.

3) *Results*: For each combination of model variant and scenario, a separate line graph was produced. Each graph contains lines for each evaluated history length trained instance which plots cumulative ADE (mean values and upper/lower quartiles) against the horizon length. These results are displayed in Figure 7.

In the flocking scenario, both model variants produced smooth, monotonically increasing ADE curves w.r.t. prediction horizon. A separation was observed between history length 1 and all other values, which formed a closely grouped cluster. Among these, history length 3 achieved the lowest cumulative ADE, while history lengths 10 and 20 produced almost identical results.

In the navigation scenario, although a similar trend was observed, the gap between history length 1 and the remaining values was significantly larger. Similar to the flocking scenario, history lengths greater than 1 produced comparable performance and there was no clear distinction between values in the range 3 to 20.

In contrast, the coupa (SDD) scenario produced a different pattern where all curves followed a non-monotonic trend. ADE initially increased up to approximately horizon 5, followed by a decrease, and then a steady linear increase for longer horizons. Across both model variants, the performance differences between history lengths were relatively small, with history length 3 consistently achieving the best performance. The model with temporal encoding exhibited slightly greater variance between history lengths, but also achieved marginally better performance in its best case.

4) *Interpretation*: The results support the hypothesis that increasing historical context improves prediction performance, particularly when comparing a history length of 1 to larger values and suggesting a small amount of temporal information is sufficient to significantly improve the model’s ability to infer agent behaviour.

The limited performance differences between history lengths greater than 3 indicate diminishing returns, supporting the hypothesis with respect to recency in observations. This effect is especially evident in the flocking scenario, where agent behaviour is primarily influenced by local interaction rules and reducing the need for long-term temporal context.

The navigation scenario exhibited a stronger dependence on temporal context, which was expected, producing a larger performance gap between history length 1 and higher values. This aligns with the literature and expectations of goal-directed behaviour where longer temporal context is required to disambiguate future trajectories.

Contrary to expectations, the model without a temporal encoder performed comparably to, and in some cases slightly better than, the proposed approach (with temporal encoding) for very short history lengths. This may suggest that the temporal encoder introduces unnecessary complexity when insufficient historical context is available.

The results from the SDD Coupa scenario differ significantly from the simulation environments. The non-

monotonic ADE curves suggest that short-term predictions are subject to higher variability, which could be caused by noise and irregularities in real-world trajectory data. The reduced sensitivity to history length may indicate that real-world behaviour is less dependent on longer historical context, or that the model is unable to exploit this context due to data sparsity or noise. This result highlights a potential limitation in generalisation from simulation to real-world data.

D. Experiment 2, Interaction Modelling

This experiment’s goal is to test how varying the connectivity in the input-graph provided to the approach can affect its ability to predict future trajectories in a physical MAS.

1) *Design*: To assess interaction modelling, the number of close neighbours each agent can influence in the message-passing was varied. Five amounts were evaluated: 0, 1, 2, 4, and 8 nearest neighbours. For each configuration, models were trained with 5 time steps of historical context and evaluated across a prediction horizons of 10 time steps; this process used the **Extended MLP** for $k = 0$ neighbours (no message-passing needed) and the **Extended GNN** for configurations with $k \geq 1$ neighbours. This was separately conducted across the football, flocking, and little (SDD) scenarios. For each simulation, 60 separate randomly initialised instances were run, each containing 10 agents and running for 250 time steps; this was done so that variation/unique topologies within each scenario could be trained on and evaluated as well as avoiding creating complete graphs $k \geq n$ as model input.

2) *Hypothesis*: The hypothesis is that a higher degree of connectivity in the input graph will improve model’s prediction performance and that this will be an ongoing trend with diminishing returns assuming that closer neighbours will provide more informative contextual information about an agent’s BDI state than more distant neighbours; this includes the model variant lacking message-passing performing worse than the proposed approach given that agents would be unable to provide contextual information to neighbours. It’s expected that the football and simple tag scenarios will experience greater improvements in performance w.r.t. connectivity in comparison to the little (SDD) scenario because the adversarial relationships in the former scenarios compared to the neutral relationships in the latter would place more importance on interactions to inform an agent’s behaviour.

3) *Results*: For each combination of scenario and evaluation metric, bar charts were produced showing prediction error as a function of graph connectivity. The metrics used were ADE, FDE, and NL-ADE, all evaluated over a prediction horizon of 10 time steps. Each bar represents the mean error with upper and lower quartiles computed over the test set. Results are displayed in Figure 8.

In the football scenario, all metrics showed minimal variation across different connectivity values. Performance remained almost constant for $k = 0, 1, 2, 4$, and 8, with no obvious trend showing improvement or degradation as connectivity increased.

In the simple tag scenario, a clear trend was observed across all metrics. Prediction error decreased consistently as

connectivity increased from $k = 0$ to $k = 8$. The most significant improvement occurred between low connectivity values, with diminishing returns at higher values. In addition, the IQR for $k = 0$ was substantially larger than for other configurations, suggesting higher variability in predictions when no interaction modelling was used.

In the Little (SDD) scenario, ADE and NL-ADE showed relatively stable mean values across connectivity levels. However, the upper bound of the IQR decreased with increasing connectivity, indicating a reduction in worst-case prediction error. FDE exhibited a more pronounced trend, with both the mean error and the upper quartile decreasing as connectivity increased, particularly at higher values of k .

Across all scenarios, NL-ADE values were consistently slightly higher than ADE, while FDE values were significantly larger, typically ranging between 1.5 to 2 times the magnitude of ADE.

4) *Interpretation:* In the simple tag scenario, the improvement in performance with increasing connectivity supports the hypothesis that modelling interactions is important in environments with tightly coupled, adversarial dynamics. Because agents directly influence each other's behaviour in a pursuit-evasion, increasing the number of neighbours allows the model to capture these dependencies more effectively. The large IQR observed at $k = 0$ further highlights the instability of predictions when interaction information is absent and reinforces the importance of message-passing in these environments.

In contrast, the football scenario showed no meaningful improvement with increasing connectivity. This suggests that simply increasing the number of static connections in the input graph is insufficient to capture complex, rapidly changing interaction dynamics present in this situation. This aligns with observations in the literature, where static interaction graphs are noted to be inadequate for modelling scenarios with dynamic and evolving relationships between agents such as sports [10]. The approach's inability to adapt the interaction topology over time may have limited its capacity to exploit additional connectivity.

The Little (SDD) scenario exhibited more subtle behaviour. While average performance (ADE & NL-ADE) remained relatively stable, improvements in FDE and reductions in the upper quartile indicate that increased connectivity helps reduce large prediction errors. This suggests that interaction modelling contributes to robustness, particularly in more complex/uncertain situations, even if it doesn't significantly improve average-case performance. The weaker dependence on connectivity compared to simulation scenarios may reflect the less structured/loosely coupled interactions present in real-world data.

Overall, these results support the hypothesis that interaction modelling is most beneficial in scenarios where agent behaviour is strongly dependent on others. However, they also highlight a limitation of the proposed approach: The use of static, proximity-based graphs may be insufficient for capturing dynamic interaction patterns in more complex environments.

V. FUTURE WORK & IMPROVEMENTS

Although the proposed approach demonstrates effectiveness in modelling aspects of physical MAS behaviour, the experimental results unveil several limitations and providing a clear direction for future work.

- The approach's ability to capture interaction dynamics between agents is sensitive to the structure of the input graph.
- Increasing historical context provides diminishing returns beyond short-term history.
- Performance differences between the synthetic and real-world datasets suggest challenges in robustness and the approach's ability to generalise.

These observations align with the key challenge areas identified in the literature, suggesting the need for more expressive interaction modelling, improved handling of temporal dependencies, and better generalisation to diverse real-world scenarios. This section outlines potential improvements to the proposed approach, focusing on model architecture, data representation, and experimental design in order to address these limitations and enhance overall performance.

A. Model Architecture

Changes to the approach's structure may be critical to improving performance w.r.t. interaction modelling, temporal modelling, generalisation and multi-modality.

Using a GNN with a different message-passing algorithm could improve the complexity of the interaction modelling. An example could be using a *Graph Attention Network* (GAT) that weights incoming messages using an attention mechanism; coupling interaction strength with learned context, as opposed to proximity, may allow for more complex and varied interaction modelling.

Alternative temporal encoder architectures may improve the approach's capacity to extract meaning from historical context beyond the short-term despite a single-pass MLP's benefits. Recurrent encoder designs (e.g. LSTMs) may enable historical features from a given time step to influence the understanding of the historical features in the next time step directly through the hidden state, improving the distant historical context's ability to contribute.

A key limitation of the existing approach is lacking a mechanism which addresses multi-modal futures. Exchanging the decoder MLP for a GMM could allow the model to produce multiple, weighted future trajectories from an individual agent's embedding. Combining these outputs with the minimal loss hypothesis during training may result in better performance in situations that contain multiple feasible future system states.

A more radical change to the approach, focused on improving generalisation, could be to utilise supervised transfer learning in place of strict supervised learning. For trajectory prediction, the model could be pre-trained on a generic task (e.g. masked trajectory recovery), where labelled data may be more abundant, and learn general trends in constructing trajectories from existing positions. These

insights could then be transferred when the model is fine-tuned on a smaller labelled dataset for physical MAS trajectory prediction, potentially speeding up training, reducing data requirements, and boosting performance.

B. Data & Experimental Design

Improving the quality and effective use of training data could contribute to maximising the potential of the existing approach. In addition, improving how model performance is evaluated may reveal key insights that are obscured by the existing implementation.

To improve generalisation from simulated to real-world scenarios, simulations must better capture the complexity of realistic multi-agent interactions. Given VMAS's noted weakness in physics, increasing physical fidelity may enhance behaviours (e.g. avoidance and cooperation), producing more realistic training data. Furthermore, replacing static controllers with learned policies via *Reinforcement Learning* (RL) could enable agents to exhibit more complex, emergent behaviours. These learned policies may better approximate real-world BDI states, helping bridge the gap between simulated and real-world behaviour prediction.

A key insight extracted from experiment 1 was that providing more consecutive historical context to the existing approach has diminishing returns on prediction performance. Given this constraint, a more productive use of limited historical context could be to stagger the time steps utilised as part of a training example. For example, using 5 previous time steps worth of information but using a step size of 5 between each useful time step would allow the model to effectively cover 25 time steps of history in a condensed form. Although the total quantity of information is reduced, the behavioural trend for an agent may be more obvious in this condensed form through eliminating consecutive time steps containing redundant context, potentially resulting in more effective temporal modelling.

The error metrics used in training, testing and experimental evaluation of the proposed approach not only impact how effectively the model can learn from its mistakes but on a project-wide scale affects what can be learnt from the experiments. In training, using MSE can be effective in trajectory regression by penalising large errors, but has a noted inability to handle multi-modal futures leading to mode collapse. Altering the decoder to produce distributions over future positions would allow us to use *Negative Log-Likelihood* (NLL) as an error metric for training; NLL allows the variance in a produced distribution to weight its impact in training, potentially producing better results in scenarios with multiple future modalities [11]. Although these are common metrics for evaluating trajectory prediction, displacement-based measures such as ADE, FDE and their variants struggle to quantify how functional or plausible a potential future trajectory is [13]. Alternative metrics (e.g. EMD, quantifying the “distance” between two distributions) could provide insights into whether the proposed approach produces realistic, structurally similar trajectories to the ground truth more effectively than point-wise measures [13].

C. Summary

Overall, the proposed improvements highlight the importance of addressing the key challenges identified throughout this project in a more integrated manner. Enhancing interaction modelling through more expressive graph representations, improving temporal encoding to better utilise historical context, and incorporating mechanisms for multi-modal prediction are all key to advancing the capability of the approach. Likewise, refining data generation and evaluation strategies is essential to improving generalisation and ensuring that model performance reflects realistic multi-agent behaviours.

Collectively, these directions suggest that effective behaviour prediction in physical MASs requires not only architectural improvements, but also a holistic consideration of data, training objectives, and evaluation methodologies. Future work that combines these elements is likely to yield more robust and scalable solutions for modelling complex multi-agent interaction dynamics.

VI. CONCLUSION

This project aimed to investigate behaviour prediction in physical MAS, framed through trajectory prediction. By identifying key challenges in the literature, a clear focus on modelling interaction dynamics and temporal dependencies was established and a graph-based neural approach was designed for this task. This approach was evaluated using synthetic data and real-world trajectory data from the Stanford Drone Dataset (SDD) as part of a complete machine learning pipeline. Overall, this report demonstrates that graph-based approaches provide an effective, flexible framework for modelling multi-agent interactions despite key limitations in static interaction representation, effective temporal modelling and generalisation between simulated and real-world scenarios. Ideally, these insights contribute to a better understanding of the challenges in multi-agent behaviour prediction and provide a foundation for future work into more adaptive, multi-modal and generalisable modelling approaches.

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